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RESEARCH

Proposal

Identifying Critical Motorcycle–Car and Car–Car Crash Scenarios under ASEAN Traffic Conditions for Future ASEAN NCAP Safety Assessment Using Computer Vision-Based Video Analysis

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TEAMS

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PROPOSAL BACKGROUND

- Motorcycles are the dominant mode of transportation across many ASEAN countries, including Indonesia, Malaysia, Thailand, and Vietnam. Despite their mobility benefits, motorcyclists remain among the most vulnerable road users and account for a substantial proportion of road traffic fatalities in the region.

- To address road safety challenges, ASEAN NCAP has continuously expanded its vehicle safety assessment framework and identified both vehicle occupant protection and motorcyclist safety as key priorities. Emerging safety technologies such as Autonomous Emergency Braking (AEB), Blind Spot Technology (BST), Rear Cross Traffic Alert/Braking (RCTA/B), and motorcycle-oriented active safety systems are expected to play an increasingly important role in future assessment protocols.

- However, ASEAN traffic conditions differ considerably from those in Europe, North America, and Japan. High motorcycle exposure, mixed traffic operations, lane filtering, close-proximity overtaking, aggressive gap acceptance, and complex interactions between motorcycles and passenger vehicles create unique crash risks that may not be adequately represented by existing safety assessment scenarios.

PROPOSAL BACKGROUND (Cont.)

- Although crash databases are widely used for road safety analysis, they typically provide limited information regarding vehicle trajectories, collision configurations, impact characteristics, and driver or rider responses immediately before a crash. Consequently, the mechanisms leading to motorcycle–car and car–car crashes under ASEAN traffic conditions remain insufficiently understood.

- The increasing availability of traffic surveillance footage and advances in computer vision technologies provide new opportunities to reconstruct crash events and extract detailed information regarding vehicle movements, pre-crash manoeuvers, impact configurations, and collision dynamics. Such evidence can improve understanding of real-world crash scenarios and support data-driven safety assessment development.

- Therefore, there is a need to systematically identify and characterize critical motorcycle–car and car–car crash scenarios using computer vision-based video analysis. The findings are expected to provide ASEAN-specific evidence for future ASEAN NCAP vehicle and motorcyclist safety assessment protocols and contribute to the development of representative crash evaluation scenarios under ASEAN traffic conditions.

PROBLEM STATEMENT

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- Previous studies have primarily focused on crash frequency analysis, injury severity modelling, driver and rider characteristics, and vehicle-related factors. However, limited studies have systematically examined how motorcycle–car and car–car crashes occur under real-world ASEAN traffic conditions using video-based crash analysis.

- Existing crash databases typically provide information on crash outcomes, locations, and injury severity but contain limited information regarding pre-crash manoeuvres, vehicle trajectories, collision configurations, impact characteristics, and driver or rider responses immediately before collision.

- Furthermore, many vehicle safety assessment scenarios currently adopted worldwide are derived from crash data collected in Europe, North America, and Japan. These scenarios may not fully represent the unique traffic conditions observed in ASEAN countries, where mixed traffic operations, high motorcycle exposure, and complex interactions between motorcycles and passenger vehicles are common.

- Therefore, there is a critical need to systematically identify and characterize motorcycle–car and car–car crash scenarios using computer vision-based video analysis in order to generate ASEAN-specific evidence and support the development of future ASEAN NCAP vehicle and motorcyclist safety assessment protocols.

RESEARCH OBJECTIVES

01

To identify and classify critical motorcycle–car and car–car crash scenarios using computer vision-based video analysis of traffic accident recordings from selected ASEAN countries.

02

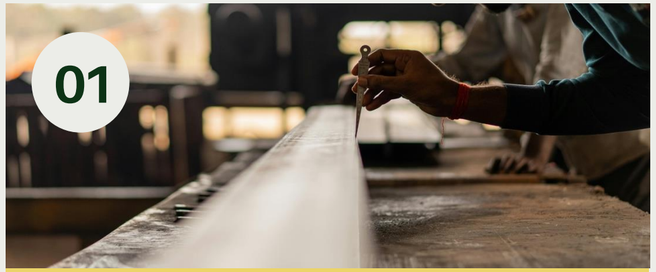
To quantify the characteristics and severity of motorcycle–car and car–car crashes using vehicle trajectories, impact speed, collision angle, collision configuration, and pre-crash manoeuvers extracted from video data.

03

To develop evidence-based ASEAN-specific crash scenario recommendations for future vehicle and motorcycle safety assessment under the ASEAN NCAP framework.

METHODOLOGY

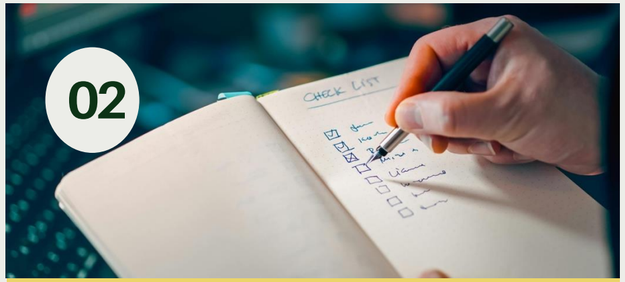
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Step 1

Crash Video Data Collection

- Collect motorcycle–car and car–car crash videos from Indonesia and Malaysia.
- Sources may include:
  - Traffic management CCTV systems
  - Police traffic surveillance recordings
  - Publicly available traffic accident videos
  - Traffic monitoring centres
- Focus on urban roads and intersections with frequent crash occurrences.



Step 2

Computer Vision-Based Crash Reconstruction

Using: YOLO, DeepSORT, Trajectory Extraction; to get:

- vehicle trajectories
- speed profiles
- relative speed
- movement patterns
- pre-crash manoeuvres
- impact location



Step 3

Crash Scenario Identification and Characterisation

1. **Motorcycle–Car**

Rear-End, Turn Across Path Oncoming, Turn Across Path Lateral, Crossing, Same Direction Turn

2. **Car–Car**

Rear-End, Crossing, Turning Collision Side Impact, Opposite Direction Collision



Step 4

ASEAN NCAP Scenario Development

- Translate representative crash scenarios into ASEAN-specific safety assessment scenarios.

MILESTONE

Deliverable 1

Crash Video Database:
- 200 motorcycle–car crash cases
- 200 car–car crash cases

Indonesia and Malaysia
Metadata:
Location, time, vehicle type, crash type

Deliverable 2

Reconstructed Crash Dataset, containing:
- trajectories
- speeds
- vehicle movements
- collision configurations
- pre-crash behaviour

Deliverable 3

Crash Scenario Database:
- **crash type**
- **collision configuration**
- **impact characteristics**
- **severity level**
- **scenario frequency**

Deliverable 4

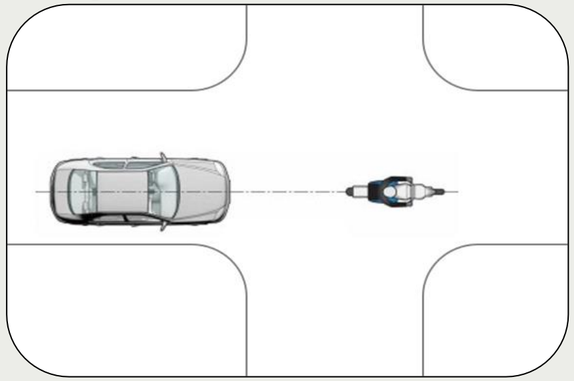
ASEAN Crash Scenario Classification:

- representative crash scenarios
- scenario frequency
- scenario severity
- ASEAN NCAP relevance

CRASH SCENARIO CLASSIFICATION

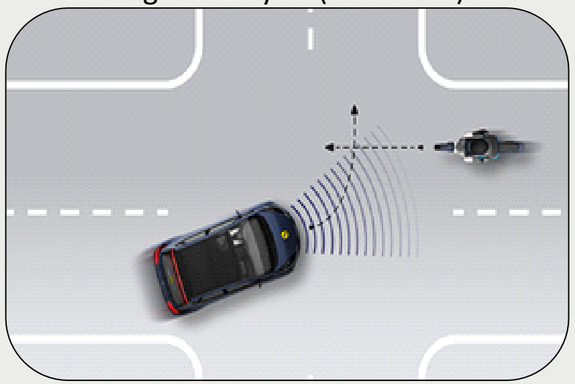
1. REAR MOVING

Rear-end conflict when both vehicle are moving in the same lane.



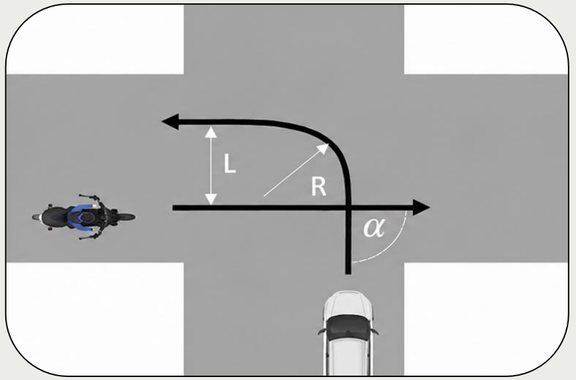
2. TURN ACROSS PATH – ONCOMING

A vehicle turns across the path of an oncoming motorcycle (vice versa).



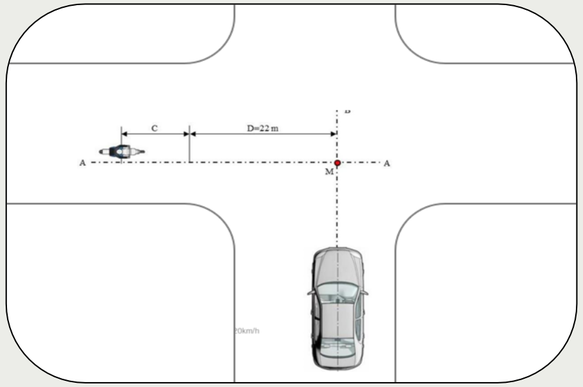
3. TURN ACROSS PATH - LATERAL

A vehicle turns across the path of a laterally approaching motorcycle (vice versa).



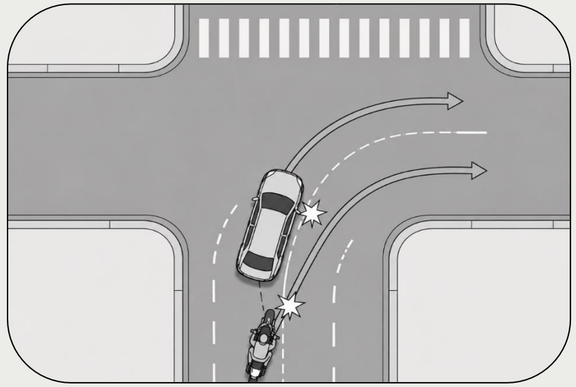
4. CROSSING

A motorcycle and a vehicle cross paths from perpendicular directions.



5. SAME DIRECTION TURN

A motorcycle and a vehicle turn simultaneously in the same direction.



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CRASH SCENARIO VARIABLES

1. Crash Configuration

- Crash Type
- Road User Pair
- Vehicle 1 Maneuver
- Vehicle 2 Maneuver

2. Crash Dynamics

- Vehicle 1 Speed
- Vehicle 2 Speed
- Relative Speed
- Impact Speed
- Impact Angle
- Evasive action

3. Severity

- Point of Impact
- Crash Severity

4. Environment variables

- Weather

PROJECT COVERAGE



Number of data

Number of cases: 400

Number of variables: 13

ASEAN Countries covered

- [V] Malaysia
- [V] Indonesia
- [ ] Thailand
- [ ] Vietnam

Research Approach

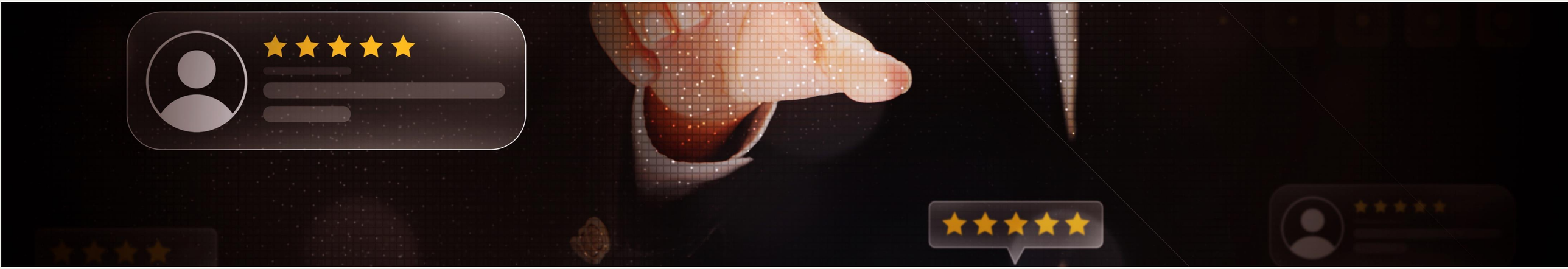
- [V] Real-world accident data analysis
- [V] Secondary data analysis
- [ ] Survey
- [ ] Development
- [ ] Simulation

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GANTT CHART

No	Activities	2026					2027									
		Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
1	Activity	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
2	Kick-off meeting & project planning	■														
3	Coordination with authorities & partners	■	■													
4	Identification of crash video sources	■	■													
5	Collection of crash videos	■	■	■	■											
6	Development of YOLO + DeepSORT pipeline	■	■	■												
7	Vehicle detection & trajectory extraction		■	■	■	■										
8	Manual validation & quality assurance			■	■	■	■									
9	Crash event identification & coding			■	■	■	■									
10	Crash scenario classification					■	■	■	■							
11	Crash reconstruction & trajectory analysis						■	■	■	■						
12	Analysis of crash characteristics						■	■	■	■	■					
13	Identification of critical crash scenarios							■	■	■	■					
14	Development of ASEAN crash scenario catalogue									■	■	■				
15	Recommendations for ASEAN NCAP										■	■				
16	Final report writing												■	■	■	
17	Journal paper preparation													■	■	
18	Final presentation & dissemination															■

PROPOSAL RECAP



Visit www.aseancap.org/ourtest for ASEAN NCAP protocol example.



No. 1-10 2026-2030 Protocols



No. 11-20 2026-2030 Protoc...

With your research, which ASEAN NCAP Protocol will benefit?

This study supports future ASEAN NCAP safety assessment development by providing ASEAN-specific evidence on motorcycle–car and car–car crash scenarios derived from computer vision-based video analysis. The findings will identify critical crash configurations, impact characteristics, and pre-crash manoeuvres to inform future AEB-MC, BST, RCTA/B, ARV, CCR, and CTR evaluation scenarios.

Thank you